...GM (genetically modified) crop.	Start: Pollution can occur in water...	...due to sewage, fertiliser or toxic chemicals	Food security is...
...having enough food to feed a population.	The rate of animal growth can be increased by...	...using high protein foods.	Conservation of fish stocks at a sustainable level is achieved by...
...control of net size and the use of fishing quotas.	Carbon dioxide is released into the atmosphere due to the burning or ..	...decay of peat.	Biological factors which are threatening food security include...
...new pests and pathogens that affect animals and crops.	Pollution can occur in air...	...from smoke and acidic gases	Efficiency of food production can be improved by reducing energy transfer from food animals to ...
... the environment by limiting their movement and controlling the temperature of their surroundings.	Biodiversity is reduced by...	...pollution killing plants and animals.	The fungus Fusarium is used for making ...
.... mycoprotein, a protein-rich vegetarian food	It is important to maintain fish stocks at a level where breeding continues to avoid...	....species disappearing completely in some areas of the sea.	Secondary consumers are...
...carnivores that eat herbivores.	Pollution can occur on land...	....from landfill and from toxic chemicals.	There is a conflict between the need for cheap compost to increase food production and...
....the need to conserve peat bogs and peatlands habitats and their biodiversity	Human insulin is produced by...	...a genetically modified bacterium. It can be used to treat people with diabetes.	When constructing a pyramid of biomass, trophic level 1 is ...

... always at the bottom.	Rapid growth in human population and an increase in standards of living mean...	... more resources are used and therefore more waste is produced.	Primary consumers are...
...herbivores which eat plants and algae (producers).	Deforestation in tropical areas has occurred to...	...provide land for cattle, rice fields and growing crops for biofuels.	Decomposers break down...
...dead plant and animal material by secreting enzymes into the environment.	Biodiversity is the...	...variety of all the different species of organisms on earth, or within an ecosystem.	Global warming is caused by...
...increasing levels of carbon dioxide and methane in the atmosphere.	Programmes to reduce the effects of humans on ecosystems and biodiversity include...	...breeding programmes for endangered species, reduction of deforestation and CO_2 emissions	Trophic level 1 contains ...
...plants and algae which make their own food and are called producers.	Losses of biomass from a food chain are because...	...some is lost as faeces and some is lost as waste e.g. CO_2 & H_2O during respiration	Apex predators are...
... carnivores with no predators.	Pyramids of biomass represent...	...the relative amount of biomass in each level of a food chain.	Scientific consensus about global warming is based on...
...systematic reviews of thousands of peer reviewed studies.	Humans reduce the amount of land available for other species by...	... building, quarrying, farming and dumping waste.	The fungus Fusarium is grown on...
...glucose syrup, in aerobic conditions.	Photosynthesis is a process carried out by green plants and algae (producers) in which...	...light is trapped to fix CO_2 and combine it with hydrogen from H_2O to make organic compounds & O_2	Golden rice is an example of a....

Teaching notes

There are 32 cards, allowing it to be used with a whole class.

The activity could also be done individually, in pairs or in small groups to produce a complete ordered set of cards.

The content of the cards enables students to revise 4.7.3 Biodiversity and the effect of human interaction on ecosystems, 4.7.4 Trophic levels in an ecosystem (biology only) and 4.7.5 Food production (biology only).